CSE221
Student ID: _____

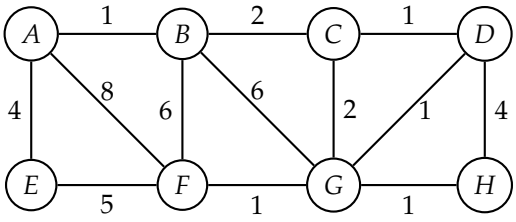
QUIZ 3
TOTAL MARKS: 10
DURATION: 15 MINUTES

**A**

## Problem 1: Dijkstra's Algorithm (8 points)

Suppose Dijkstra's algorithm is run on the following graph, starting at node $B$.

Fill in the following tables with the contents of the priority queue, which node is popped during each iteration, and the final $\texttt{dist}(u)$ values. Break ties alphabetically. The first one has been done for you.




| Iteration | Priority Queue | Node Popped |
|---|---|---|
| 0 | $A(\infty), B(0), C(\infty), D(\infty), E(\infty), F(\infty), G(\infty), H(\infty)$ | - |
| 1 | A(1), C(2), D(∞), E(∞), F(6), G(6), H(∞) | $B$ |
| 2 | C(2), D(∞), E(4), F(6), G(6), H(∞) | A |
| 3 | D(3), E(4), F(6), G(5), H(∞) | C |
| 4 | E(4), F(6), G(4), H(7) | D |
| 5 | F(6), G(4), H(7) | E |
| 6 | F(5), H(5) | G |
| 7 | H(5) | F |
| 8 | | H |

| $u$ | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ | $G$ | $H$ |
|---|---|---|---|---|---|---|---|---|
| $\texttt{dist}(u)$ | 1 | 0 | 2 | 3 | 4 | 5 | 4 | 5 |